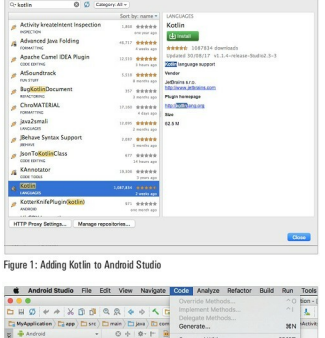



Figure 1: Adding Kotlin to Android Studio

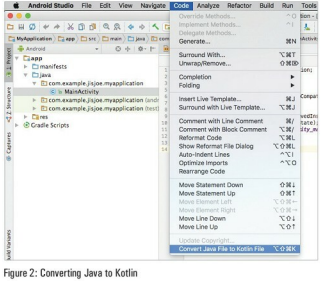



Figure 2: Converting Java to Kotlin

select *File > New > New Project.* The slide asking for the project's name will have an option underneath, *'Include Kotlin Support'.* Checking the box will enable Kotlin, and this is the starting activity — the MainActivity (default name) named *MainActivity.kt*. For other versions, the user needs to convert the existing Java class to Kotlin, by opening the *MainActivity.java* file, and then selecting *Code > Convert Java file to Kotlin file* (Figure 2).

Figure 3 shows the differences between Java class and Kotlin class. The following differences can be noted:

1. The extension has changed from *.Java* to *.kt*.
2. *extends AppCompatActivity* is changed to: *AppCompatActivity()*. In Kotlin, the declaration of a variable name is followed by a ':', after which the datatype is specified.
3. 'override' becomes part of the *onCreate* function and the function syntax is changed to 'fun onCreate'




Figure 3: Differences between the Java and the Kotlin class

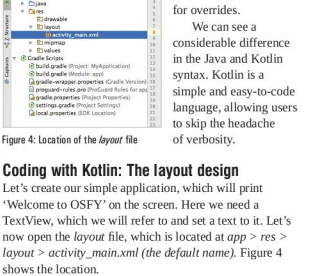



Figure 4: Location of the *layout* file

in Kotlin. Unlike Java, Kotlin requires explicit annotations for overridable members for overrides.

We can see a considerable difference in the Java and Kotlin syntax. Kotlin is a simple and easy-to-code language, allowing users to skip the headache of verbosity.

## Coding with Kotlin: The layout design

Let's create our simple application, which will print 'Welcome to OSFY' on the screen. Here we need a TextView, which we will refer to and set a text to it. Let's now open the *layout* file, which is located at *app > res > layout > activity_main.xml (the default name).* Figure 4 shows the location.

Here, we can see that a TextView has already been added to the *layout* file. Switch to the 'Design' section from the TextView (this can be changed at the bottom part of the displayed window, containing code). Next, select the TextView from the Component Tree window (Figure 5).

In Figure 5, on the left, we have the *Palette* window, where we can select and add more widgets to the application screen. To the right, we have the *Properties* window, where we can make property changes to the selected TextView.

The most important property is the ID, so provide a suitable name, or rather, set the ID value as 'textview' itself for easy identification. This ID is used to uniquely identify the particular element in our Java/Kotlin code. We can

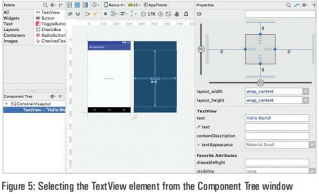



Figure 5: Selecting the TextView element from the Component Tree window